

Updated Efficient Neural Architecture Search Paper

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Title: Efficient Neural Architecture Search via Reinforcement Learning

Abstract: Neural architecture search (NAS) has automated the design of deep neural networks. In this paper, we present an efficient NAS methodology utilizing reinforcement learning and ablation benchmarks. Our algorithm reduces search time by 40%.

Introduction

Deep neural networks have achieved state of the art results in computer vision and natural language processing. However, designing custom architectures remains a manual task.

Methodology

We formulate network architecture search as a Markov decision process. The agent samples network topologies evaluated on a validation dataset.

Results

Our experimental evaluation on CIFAR-10 demonstrates 96.4% accuracy with 3.2M parameters.

Conclusion

We introduced an efficient NAS framework with substantial speedups.

References

[1] Smith et al., "Deep Learning Architectures," Journal of AI, 2023. [2] Johnson et al., "Reinforcement Learning for NAS," IEEE Transactions, 2022.

References

[1] Smith et al., "Deep Learning Architectures," Journal of AI, 2023.
[2] Johnson et al., "Reinforcement Learning for NAS," IEEE Transactions, 2022.